	Player Category	Player Example	Player Roles	Player Roles Details
Primary Project Maintainers	Internet Infrastructure	• Network service providers	• [DISCOVER] Identify general infrastructure vulnerabilities	• Detect insecure con • Identify exposed or • Log vulnerabilities • Share general vulne
			• [DISCOVER] Detect general infrastructure threats	• Detect high-volume Monitor for route h • Identify mass-scann • Share threat intell
			• [REMEDIATE] Address infrastructure vulnera	• Apply patches or co onfigurations, expo

			bilities and respond to detected threats	• Respond to active t oling, rate-limitin • Isolate compromised • Coordinate with inc ats. • Notify affected dow
			• [REPORT] Share identified infrastructure vulnerabilities with relevant stakeholders	• Report discovered v rs and affected sec • Notify CERTs, ISACs, • Contribute vulnerab
			• [REPORT] Share observed network-level threats with relevant stakeholders	• Share threat intell • Report incidents of or wallet access. • Inform infrastructu s that may degrade • Escalate emerging t broader disseminati
			• [ASSESS] Assess risks based on shared v	• Analyze disclosed v e availability for

			ulnerabilities and threats int elligence	• Review threat repor nfrastructure used b • Coordinate internal c risks. • Update risk models a ould propagate acro • Revise internal bla elligence.
			• [CONTRIBUTE] Develop and maintain general i nternet security and reliabili ty standards	• Participate in techn ting, DNS, and peer • Contribute operatio nd network abuse. • Support efforts to a • Promote adoption of • Ensure that general stries, including b
			• [USE STANDARD] Apply general internet infrast ructure standards for secure a nd resilient network operation	• Implement routing s ike MANRS. • Use standardized DN • Apply incident resp

				s	standards. • Maintain infrastruc ETF, ICANN, or regi • Support interoperab luding blockchain a
	Blockchain Protocol	**Blockchain**	• L1 Blockchain foundati on (Ethereum Foundatio n, Solana Foundation, etc) • L2 Blockchain foundati on (Arbitrum Foundatio n, etc) • Node Providers (incl R PC/Archive Nodes) • Other Open-Source Serv ice Providers (node so ftware, etc)	• [DISCOVER] Undercover vulnerabilities in consensus, client code, or inf rastructure	• Conduct internal au plementation. • Analyze the behavio level flaws. • Monitor community b rabilities. • Engage in fuzzing, e transitions and s
				• [DISCOVER] Detect network anomalies or pr otocol-level threat	• Detect abnormal val itation. • Monitor fork attemp • Identify attack pat p). • Track infrastructur ayers.
				• [REMEDIATE] Develop and release protocol u pgrades	• Design, implement, ss identified vulne • Coordinate upgrade • Execute emergency u • Conduct regression • Communicate upgrade app developers.

				• [REPORT] Share protocol or node vulnera bilities to stakeholders	• Disclose confirmed ers through respons • Publish detailed se hes. • Share vulnerability ocol implications e • Collaborate with up
				• [REPORT] Disseminate threat intelligenc e on consensus or infrastructu re attacks	• Disseminate informa ttacks) to the comm • Issue alerts about perations. • Coordinate with exc ence. • Report security inc iscord, GitHub, Twi

				• [ASSESS] Assess risks using shared thre at and vulnerability data	• Evaluate the impact tocol performance a • Cross-reference sha n-chain logs and te • Update risk dashboa r coordinated threa • Revise internal bla elligence.
				• [CONTRIBUTE] Contribute protocol-level insi ghts to global standardization efforts	• Propose and co-auth • Participate in work vernance standards. • Align protocol beha • Collaborate with st design scalable, se
				• [USE STANDARD] Implement and uphold security and protocol standards	• Implement protocol • Align node software • Maintain backward a • Ensure security and elines.

		DeFi	• DeFi Developers • Stablecoin Issuers	• [DISCOVER] Identify security gaps in DeFi logic, contracts, or integrati ons	• Conduct internal au atterns to detect v • Monitor economic bel on) for protocol de • Run bug bounty prog ry.
				• [DISCOVER] Identify exploit patterns and attack vectors in DeFi systems	• Monitor abnormal us loits. • Track malicious gov pricing. • Detect smart contra
				• [REMEDIATE] Deploy security fixes and prot ocol upgrades	• Implement emergency eatures) in respons • Execute governance ation patches. • Publish upgrade doc • Collaborate with mu ut of changes.
				• [REPORT] Report security flaws in DeFi protocols to stakeholders	• Report discovered v ners or upstream pr • Notify liquidity pr ublic disclosure. • Publish technical p

				• [REPORT] Report detected threats to sta keholders across the DeFi ecos ystem	• Alert peer protocol pts. • Report scam contrac • Coordinate across e
				• [ASSESS] Monitor DeFi vulnerabilities i nformed by ecosystem disclosur es	• Review threat repor firms to assess pot • Cross-reference sha lnerabilities) with • Monitor ecosystem-w t similar exposure • Analyze disclosed i (e.g., collateral r • Collaborate with ec s) to evaluate casc • Revise internal bla

					elligence.
				• [CONTRIBUTE] Collaborate on open standards for DeFi security and function ality	• Propose new pattern oracle usage in for • Co-develop secure p • Share security best tandards.
				• [USE STANDARD] Adopt recognized standards for DeFi contracts and integration s	• Implement controls • Follow formal verif egulatory guidance. • Ensure off-chain in e and deployment st
	Blockchain Network	Network Operator	• Miners • Validators (Proposers / Builders) • Mining/Validation Pool Operators • Sequencers / Data Mana gers • Relayers	• [DISCOVER] Identify weaknesses in network infrastructure or protocol beh avior	• Monitor block produ eal protocol or cli • Observe validator o ate flaws in consen • Identify abnormal p tted to the network. • Detect vulnerabilit ased manipulation, • Flag infrastructure that point to deepe

				• [DISCOVER] Detect network-level threats a nd suspicious activity	• Monitor for malicio double-signing. • Detect suspicious t ontract deployment). • Identify attempts t rder. • Observe anomalies i te targeted attacks. • Flag data availabil
				• [REMEDIATE] Take immediate action to apply security fixes and mitigate th reats	• Upgrade client soft • Temporarily pause o ropagation. • Follow coordinated g live threats. • Halt validation or detected. • Apply real-time mit plate settings. • Ensure continued up breakdown.
				• [REPORT] Report observed vulnerabilitie s in network infrastructure or protocol behavior	• Report bugs or faul dation errors) to c • Alert infrastructur in data availabilit • Coordinate responsi itical actors are i

				• [REPORT] Report observed threats to the network or ecosystem	• Alert core developer empts. • Report observed MEV onsensus safety. • Share intelligence gs. • Notify relayer or b
				• [ASSESS] Evaluate risks to the network using shared threat and vulner ability intelligence	• Review disclosed vu to assess potential • Monitor node behavi sactions, or MEV ab • Compare shared atta de activity and mem • Adjust validator or able feature). • Revise internal bla elligence.

				• [CONTRIBUTE] Help shape operational and sec urity standards through networ k participation	• Provide feedback on dator or miner beha • Participate in test scussions, IETF/CCS • Support development ly and what doesn't • Share lessons learn les, or fee market • Help validate the f and reporting outco
				• [USE STANDARD] Operate in alignment with esta blished protocol and security standards	• Participate in on-c security implicatio • Run governance clie ts (e.g., in protoc • Review and evaluate ntial risks or unin • Monitor testnet dep des before mainnet
		Data Provider	• Blockchain Explorers • Indexers	• [DISCOVER] Identify anomalies and emergin	• Operate monitoring misbehavior.

				g threats from on-chain data	• Track network laten ographies. • Detect signs of cen k content. • Monitor for validat e, eclipse, or equi • For indexers: detec logic.
				• [REPORT] Share structured threat intell igence and transparency data	• Share real-time ale reats are observed. • Publish reports on otocol security or • Notify core dev tea ols. • Coordinate with pee (e.g., “war rooms”
				• [CONTRIBUTE] Help standardize/structure blo ckchain data for use across th e ecosystem	• Help define schema t clustering, event • Support or co-devel d regulators. • Contribute to trans nitoring tools.

	Blockchain App	Self-Custdodial	• Self-Custodial Wallet Providers • Transaction Managers (ex. Safe, Ginco) • Other Hardware Provide rs	• [DISCOVER] Detect security flaws in walle t software, hardware, or trans action infrastructure	• Audit wallet code a alculation. • Identify UI vulneral s, approval tricker • Analyze firmware an hysical access resi • Detect misuse of in traction or MPC) in • Monitor bug bounty abilities.
				• [DISCOVER] Detect active threats targetin g wallet users, interfaces, or infrastructure	• Monitor for phishin s. • Detect scam contrac prompts. • Identify malware an ails, or inject cod • Track emerging atta tricks, or social e • Analyze telemetry o ipulation. • Watch for firmware erabilities.
				• [REMEDIATE] Patch security flaws and stren gthen user protection mechanis ms	• Deploy software or ry mechanisms, or d • Improve user-facing approval risk alert • Enhance logic for d ns, or account drai • Integrate optional

					or optional integra ● Refactor transactio ay attacks. ● Collaborate with op oss wallet ecosyste
				● [REPORT] Report discovered security vul nerabilities to relevant stake holders	● Coordinate responsi iers, or library ma ● Notify affected eco issue may impact sh ● Submit reports to p tructure. ● Share anonymized vu tion channels. ● Publish security ad r software updates). ● Work with bug bount fied and disclosed
				● [REPORT] Share detected threats targeti ng wallet users or infrastruct ure	● Report phishing sit iders, and security ● Alert other ecosyst drainer patterns, o ● Share intelligence ased rapid response ● Publish security bu aviors. ● Contribute to black ature spoofing atte

					• Coordinate with har xtensions, or suppl
				• [ASSESS] Evaluate security exposure bas ed on disclosed vulnerabilitie s and threat intelligence	• Review vulnerabilit • Compare shared atta xisting wallet flow • Evaluate dependenci rabilities. • Conduct internal au tems are at risk. • Adjust threat model ng in the ecosystem • Revise internal bla elligence.
				• [CONTRIBUTE] Help define standards for secu re wallet operations and user protection	• Participate in work • Help define hardwar integrity, and tamp • Collaborate with pr ds for secure walle

				• [USE STANDARD] Implement recognized security and interoperability standards	• Implement widely ac • Comply with hardwar • Adopt transaction s • Ensure secure inter interfaces. • Regularly update im protocols or govern
		DApp Developers	• DApp Frontend Develope rs	• [DISCOVER] Detect vulnerabilities in smar t contract interactions and fr ontend components	• Audit frontend code on, spoofed prompts • Review smart contra fe approvals). • Test for manipulati in signing processe • Detect inconsistenc e malicious actions • Monitor user feedba ses. • Assess the use of t ration or data disp
				• [DISCOVER] Detect active threats targetin g dApp users or interfaces	• Monitor for phishin ce. • Detect malicious sm en impersonation). • Identify injection • Observe abnormal tr attack scripts. • Track external thre rontend. • Analyze user-report

					ms.
				• [REMEDIATE] Patch security flaw and enhance user protection against abuse	• Patch frontend code et interactions. • Refactor smart cont ction calls, or mal • Implement transacti sted contract verif • Integrate blocklist d or suspicious add • Improve on-chain an ompliance analysis.
				• [REPORT] Share vulnerabilities in dApp interfaces or integrations to the ecosystem	• Disclose frontend v internal teams and • Report issues in th ce maintainers or v • Coordinate responsi col teams. • Submit vulnerabilit orums. • Notify integrators plications. • Publish post-mortem akes.

				• [REPORT] Share observed threats impacti ng dApp users or infrastructur e	• Alert security comm s, or drainer behav • Report threats rela stakeholders. • Share real-time thr scord channels). • Publish security bu reats. • Contribute to threa mpersonating the dA • Coordinate with bro ge user groups. • Educate users with ssages, verify cont
				• [ASSESS] Evaluate dApp security exposur e using shared threat and vuln erability intelligence	• Review disclosed vu act on dApp operati • Cross-reference att acts and external d • Evaluate the risk o ken display logic. • Monitor ecosystem-w spoofed or misused. • Simulate attack sce r known threat cond • Use findings to inf ommunication with u • Revise internal bla

					elligence.
				• [CONTRIBUTE] Help define secure and interop erable dApp design standards	• Propose improvement ys, or multisig UX. • Contribute design f action confirmation, • Support standardiza cross-chain dApp co • Share incident resp actices.
				• [USE STANDARD] Implement established standard s for secure and interoperable dApp development	• Implement interface • Adopt wallet connec • Apply security guid alicious contract d • Ensure compatibilit ndard component lib • Continuously update
		Cutodial, VASPs	• Centralized Exchanges • Custodial Wallet Provi ders • Other Custodial Servic e Providers • Crypto Payment Process ors	• [DISCOVER] Detect vulnerabilities in cust odial infrastructure and servi ce architecture	• Conduct regular sec transaction engines. • Identify weaknesses • Test internal syste al validation logic. • Review API endpoint • Monitor operational rrors, resource exh

					• Use threat modeling ation flows.
				• [DISCOVER] Detect active threats targetin g custodial systems	• Monitor for phishin • Detect suspicious u ansaction behavior. • Identify external a • Track evolving tact campaigns. • Analyze blockchain ers, or exploit-lin • Integrate threat in coordinated threat
				• [DISCOVER] Detect vulnerabilities and thr eats linked to third-party dep endencies	• Assess third-party • Track threat intell nts. • Perform third-party ncies, or exposed c • Simulate attack pat oracle manipulation • Collaborate with ve threats if needed.
				• [ALERT] Notify core protocol or projec t maintainers of identified vu	• Report vulnerabilit ainers or foundatio • Provide detailed te

				lnerabilities or threats	gation and remediat ● Engage through offi ounty platforms. ● Flag coordinated th hain attacks). ● Follow up to suppor
				● [REMEDIATE] Patch security flaws and stren gthen AML/CFT controls	● Apply software and s, and authenticati ● Update internal con ory expectations. ● Implement secure ke ess restrictions. ● Strengthen AML/CFT ns enforcement. ● Revise user onboard ors. ● Collaborate with le ical and regulatory
				● [ESCALATE] Escalate unresolved third-part y vulnerabilities to relevant stakeholders	● Formally document a col teams. ● Notify affected eco anding risk. ● Escalate the issue ● Engage with regulat protection risks. ● Share limited publi minimizing exploita ● Update internal ven d.

				• [REPORT] Report discovered vulnerabilit ies in custodial systems to re levant stakeholders	• Report critical vul curity) to internal • Coordinate responsi ce maintainers if t • Share relevant find le. • Notify regulators o y have affected cus • Collaborate with bu ues. • Publish post-remedi lved risks and best
				• [REPORT] Share threat intelligence rela ted to attacks, fraud, and sus picious behavior	• Report observed thr nal investigators. • Share indicators of th threat intellige • Alert integrated se eats (e.g., malware • Notify law enforcem tempts, or fraud ri • Participate in cros arly warning system • Contribute anonymiz stem resilience.
				• [ASSESS] Evaluate internal exposure in	• Analyze threat inte ucture or services.

				response to external threat an d vulnerability intelligence	• Assess whether disc ld affect internal • Cross-check interna or behavioral indic • Evaluate the exposu t threat landscape. • Conduct internal re ectors. • Use findings to upd ent response where • Revise internal bla elligence.
				• [ASSESS] Monitor third-party exposure u sing shared vulnerability and threat intelligence	• Evaluate the risk i • Track threat intell ity providers. • Monitor integration n threats. • Cross-reference thi eports. • Update vendor risk • Coordinate with ser h contractual or go

				• [CONTRIBUTE] Help shape industry standards for secure custodial operation s and regulatory alignment	• Participate in indu nance) to shape sec • Propose improvement dards based on oper • Collaborate with re risk management and • Contribute to open- ge, MPC, and HSM im • Share lessons learn ne across the indus
				• [USE STANDARD] Implement and uphold recognize d security and compliance stan dards	• Adopt custody and w • Implement regulator and recordkeeping p • Align infrastructur agement (e.g., MPC/ • Integrate risk mana audit obligations. • Regularly update sy y guidance. • Participate in cert ility to users and
Affiliated Th ird Parties	Logistics and Supply Chain Consumers / Relying Party		• Traditional Finance op erators • Financial Institutions • Marketplaces • Investors / Consumers	• [ASSESS] Monitor dependency risks acros s blockchain and internet serv ice providers	• Evaluate the securi ode operators, and • Assess exposure to user-facing service • Review threat intel e internal risk ass • Incorporate blockch ent frameworks.

				• Monitor service-leve iance failure, or ca • Adapt procurement an d-party blockchain p
			• [ANALYZE] Assess third-party contractual and operational risk	• Analyze shared secu to assess contract c • Map disclosed threat evaluate vendor rel • Review whether thir ional damage. • Coordinate with pro ontractual amendmen • Assess concentratio nerable third-party • Use findings to inf meworks.
			• [SHARE] Communicate contract-related r isk findings to third-party bl ockchain protocols/apps	• Feed analysis resul e partners. • Share risk findings equate incident resp • Notify vendors of c audit clauses, data • Escalate material r otocol instability, • Use contractual cha risk positions and • Coordinate disclosu s, or exit strategi • Promote transparenc

				o specific contract
			• [CONTRIBUTE] Provide user-driven input to t he development of blockchain s ecurity standards	• Participate in indu • Contribute insights t expectations. • Advocate for securi gnment, and inciden • Support development tion, and monitorin • Push for inclusion ce or trust framewo • Validate whether pr scale consumer plat
			• [USE STANDARD] Adopt established standards in evaluating and engaging blockc hain service providers	• Apply technical and Require third-party transaction integri • Align internal risk osure, incident rep • Adopt regulatory st ncial integrity whe • Incorporate best pr perability across b • Update compliance p tations and regulat
		• Security Practitioners	• [DISCOVER]	• Perform vulnerabili namic analysis tech

	Security Expert	• Ethical hackers • Individual security ex perts • Web3 security audit fi rms • Security Researchers	Detect security flaws in block chain protocols, apps, and net works	• Conduct penetration ructure. • Use fuzzing, formal interactions. • Analyze dependency for systemic weakne • Submit findings via ate remediation. • Continuously monito ched vulnerabilitie
			• [DISCOVER] Detect emerging threats target ing blockchain systems and use rs	• Monitor blockchain anipulation. • Identify attacker b signature abuse. • Reverse-engineer ma ses. • Track threat actors urce intelligence. • Detect novel attack s. • Collaborate with in a coordinated manne
			• [ALERT]	• Initiate responsibl designated security

			Notify affected project mainta iners of identified vulnerabil ities or threats	• Alert protocol deve er attacks) are det • Provide technical d reproduce, and fix • Coordinate disclosu ies, to allow safe • Engage with vendor retesting.
			• [ESCALATE] Disclose unresolved vulnerabil ities or threats when vendors fail to remediate	• Track remediation s ty risks. • Follow responsible in accordance with • Report unremediated pto-ISACs, bug boun • Warn affected users to funds, data, or • Publish post-disclo nd respond to unres • Support broader cal cement, or governan
			• [REPORT] Disclose identified vulnerabil ities to the broader ecosystem through trusted channels	• Submit verified vul c security forums. • Publish vulnerabili s and mitigation gu

				• Contribute anonymiz • Engage in postmorte extualize the vulne
			• [REPORT] Share actionable threat intell igence with the broader blockc hain ecosystem	• Report indicators o and exploit toolkit • Share threat report (tactics, technique • Contribute intellig and community alert • Alert peers to new • Participate in real ribution, and mitig

			• [ASSESS] Evaluate security risks across blockchain systems using disco vered or reported vulnerabilit y or threat information	• Perform contextual closed or observed • Evaluate potential ivilege escalation, • Correlate threat in flaws, access contr • Publish risk analys • Update risk models with other security
			• [ANALYZE] Examine vulnerabilities and th reats to produce actionable in sights on cybersecurity for th e blockchain ecosystem	• Perform in-depth an exploit mechanics, • Aggregate findings codebase weaknesses, • Publish threat land teams, developers, • Develop tools, chec nerabilities.

			• [SHARE] Disseminate analytical finding s and on vulnerabilities and t hreats to inform and strengthe n the blockchain ecosystem	• Publish technical a impact of discovere • Share synthesized t mic papers to raise • Contribute to publi m security incident • Collaborate with de nd widely dissemina • Translate findings and vendor selectio • Facilitate learning projects or ecosyst
			• [CONTRIBUTE] Provide expert input to securi ty standard development for bl ockchain systems	• Participate in stan closure, key manage • Draft or review pro es and threat resea • Recommend technical blockchain systems. • Promote alignment b ect actual develope
			• [PROMOTE STANDARD]	• Develop auditing to level security stan

			Build tools and frameworks tha t embed and operationalize sec urity standards	• Create developer fra ized standards. • Embed standards int iance checks. • Maintain open-sourc loyment, node confi • Collaborate with pr g (e.g., SDKs, depl • Educate developers ture aligned with i
	Auditor	• Auditors • Audit firms	• [DISCOVER] Detect security weaknesses in client systems through formal audits	• Conduct structured rastructure. • Perform code review ntrol issues, and u • Simulate real-world es. • Continuously update blockchain ecosyste
			• [DISCOVER] Detect security threats releva nt to client systems and opera tions	• Identify relevant t pp, or infrastructu • Assess exposure to nipulation, or gove • Analyze client arch es may introduce ri • Perform red-team si nses.

			• [ALERT] Notify clients of identified s ecurity vulnerabilities and po tential threats	• Deliver comprehensi • Schedule debrief se before deployment o • Provide remediation ject timeline. • Maintain secure com ct client interests
			• [ESCALATE] Initiate responsible disclosur e when critical vulnerabilitie s or threats that could impact the broader ecosystem are not remediated	• Track the remediati imelines. • Initiate escalation to project stakehol ERTs, Crypto-ISAC). • Coordinate with leg ligations and profe • Provide anonymized d systemic risks wi
			• [REPORT] Alert clients and relevant sta keholders about potential thre ats identified	• Notify clients of s tors uncovered duri • Identify and flag p • Recommend immediate tegrations based on • Share anonymized in Ts, when threats po • Collaborate with af emediation.

			• [ASSESS] Conduct security audits and pr ovide advisory services to cli ents	• Perform security au s, and node infrast • Evaluate cryptograp l security measures. • Conduct both manual • Test for known vuln ess business logic • Provide audit repor on steps. • Follow up with re-a
			• [ANALYZE] Examine trends in vulnerabilit ies and threats to improve aud it practices	• Aggregate findings patterns, or protoc • Analyze on-chain ex ve test coverage. • Review threat intel ith real-world risk
			• [SHARE] Publish insights and updates t o audit methodologies	• Publish technical a bilities and how th

				• Update internal aud ts are more robust • Release updates to es, encouraging con • Present findings at ge exchange and com • Create educational t frameworks and su
			• [CONTRIBUTE] Offer expert input into the de velopment of security and audi ting standards	• Join working groups • Provide feedback on ations, or incident • Translate field exp • Advocate for standa hain layers.
			• [PROMOTE STANDARD] Align audit frameworks with ev olving standards	• Integrate technical • Regularly review ne ology, scoring, or • Educate clients on gn and deployment s • Collaborate with st consistently applie
	Insurance Company	• Insurance Company (Cyb er Insurance)	• [ASSESS] Evaluate cybersecurity risks a nd underwrite blockchain-relat ed insurance coverage	• Conduct security ri readiness, and thir • Evaluate client adh t, off-chain infras • Define coverage par sks, attack history,

				• Provide tiered poli audit results. • Collaborate with br dator slashing, bri • Use breach data and lity criteria.
			• [CONTRIBUTE] Help shape cybersecurity and r isk standards through insuranc e industry expertise	• Participate in stan security and disclo • Advise on risk quan nt response maturit • Contribute incident kchain services. • Validate proposed s world insurance por • Bridge gaps between oss risk, compliance
			• [PROMOTE STANDARD] Reflect standards on cyber ins urance coverage and pricing	• Offer premium disco pendent audits. • Incorporate securit • Reward proactive th ough more favorable • Adjust pricing mode ignals, and validat • Use standards as a imited operational
			• [DISCOVER]	• Continuously monito ress attribution te

	Blockchain Analytics Firm	• Blockchain Analytics firms (Chainalysis, Elliptic, etc)	Monitor blockchain networks to detect illicit activity and threat patterns	• Identify behavioral loit laundering. • Flag addresses or c mised protocols. • Integrate off-chain on-chain behavior a • Provide clients wit based on real-time
			• [REPORT] Notify clients and the ecosystem of potential threats based on detected anomalies or illicit behavior	• Send real-time aler patterns emerge. • Notify clients (blo ed through their pl • Provide actionable • Alert compliance te • Offer contextual ri
			• [ASSESS] Provide advisory services and forensic support to clients	• Conduct wallet trac or law enforcement • Help institutions a • Provide compliance behavior analysis. • Assist in internal hes.

				• Support pre-listing forms.
			• [ANALYZE] Continuously examine threat tr ends to improve monitoring sys tems and inform clients	• Track how launderin • Analyze smart contr • Develop behavioral • Feed analytics resu gorithms. • Brief clients on ec s.
			• [SHARE] Disseminate aggregated threat intelligence and forensic insi ghts to the ecosystem	• Publish quarterly o atterns, and laund • Share wallet blackl law enforcement. • Contribute to multi es. • Offer workshops or orensic learnings. • Maintain open APIs
			• [CONTRIBUTE] Participate in development of risk classification and compli ance standards	• Provide expert inpu rameworks. • Help define categor ctivity detection.

				• Recommend best prac ion protocols. • Support interoperab • Contribute to the d
			• [PROMOTE STANDARD] Embed accepted standards into monitoring tools and client-fa cing products	• Implement FATF-alig ards. • Embed industry-appr nto client-facing s • Customize alerts an ctations. • Provide automated r • Ensure traceability, h transparency stan
Independent / Public Third Parties	Certification Body	• Security Certification schemes • Testing/Certifying Lab (CC, CMVP)	• [ASSESS] Evaluate security posture and issue certifications based on defined standards	• Perform structured platforms against e • Use repeatable test t, access controls, • Analyze system-spec party integrations, • Issue certification ps are identified. • Maintain audit trai in compliant over t • Support public tran awarded.
			• [CONTRIBUTE] Contribute to standards	• Participate in stan urity, custody assu

				• Offer feedback base iently tested, or m • Propose new criteri • Help align terminol hemes to reduce dup • Contribute anonymiz models and threat r
			• [PROMOTE STANDARD] Integrate standards into certi fication criteria	• Map industry-recogn certification. • Update certificatio r consensus-driven • Define pass/fail th t standards across • Train assessors and ness in evaluations. • Encourage adoption r applicants prepar
	Academia	• Academia (Academic Pee r-Reviewed Papers) • Contribute to standard developmentAcademia (S ponsored Research)	• [ANALYZE] Conduct academic research on s ecurity vulnerabilities and em erging threats	• Design and conduct aws, validator beha • Develop formal mode ics, and adversaria • Investigate real-wo emic risks. • Publish peer-review knowledge on blockc • Collaborate with in issues • Provide an academic kchain ecosystem.

			• [SHARE] Disseminate research findings and build capacity through edu cation and public engagement	• Deliver lectures, s raphy, and protocol • Publish research in en reach beyond aca • Organize or contrib ent, vulnerability • Mentor students and practical security • Participate in mult engineering practic • Encourage open dial ent and co-creation
			• [CONTRIBUTE] Shape the development of techn ical and policy standards thro ugh research-based insights	• Participate in tech iew blockchain-rela • Contribute empirica marks and control d • Review or co-author l or theoretical fi • Advocate for standa eness. • Facilitate alignmen d practical standar • Act as a neutral vo ds reflect both fea
	Law Enforcement Authorities		• [ASSESS]	• Investigate crimes n involving digital

		• Law Enforcement Author ities (Europol, INTERP OL, FBI, etc)	Conduct investigations and tak e enforcement action based on gathered evidence	• Use blockchain anal flows and identify • Arrest individuals ctional requirement • Coordinate with pro • Protect victims and
			• [ANALYZE] Interpret threat intelligence to determine enforcement strat egy and legal pathways	• Review threat repor ss credibility. • Cross-reference wal l conduct. • Determine whether d gal frameworks (e.g. • Assign investigativ g) based on case ch • Consider diplomatic
			• [SHARE] Disseminate enforcement strate gies and lessons learned to su pport ecosystem security	• Financial

			• [CONTRIBUTE] Help shape international and n ational AML/CFT standards thro ugh investigative insight	• Collaborate with st sion to provide fee • Recommend threshold privacy coins, NFTs, • Contribute anonymiz ation techniques. • Support efforts to cement data-sharing • Align law enforceme reporting and super
			• [PROMOTE STANDARD] Integrate standard into enforc ement	• Incorporate interna virtual assets. • Align investigative l interoperability • Apply technical sta when analyzing on-c • Use standards-based the classification • Collaborate with co supports industry a • Train officers and dards within blockc
	Financial Regulatory Authoriti es	• Financial Regulatory A uthorities (SEC, CFTC,	• [ASSESS] Evaluate risks of blockchain-r elated projects from the lens	• Monitor blockchain at may result in in

		JFSA, MAS, FCA, BaFin, AMF, etc) Courthouse and attorne y (for trial)	of customer protection, crime prevention, and financial stab ility	• Assess the impact o coin depegging-on c • Evaluate business m ntial exposure to f • Respond to reports gatory processes. • Engage internal or itigation involving
			• [ANALYZE] Conduct cross-sectoral risk an alysis of blockchain market tr ends and systemic vulnerabilit ies	• Aggregate incident oviders to evaluate • Analyze network str points of failure o • Assess the resilien idge failure, insol • Map illicit activit er enforcement, and • Produce internal or licy perspectives.
			• [SHARE] Share recommendations based on analysis	• Release public stat tokenization, staki • Engage in bilateral responses and enfor • Publish lessons lea r supervisory asses

				• Provide technical r er protection mecha • Conduct outreach se egulatory positions.
			• [CONTRIBUTE] Help shape international and n ational financial regulatory s tandards based on analysis	• Participate in stan olicy frameworks. • Propose revisions t supervision, or tok • Incorporate courtro digital assets. • Recommend regulator tion, and financial • Support coordinatio ght and prosecution
			• [PROMOTE STANDARD] Integrate standard into regula tions	• Codify internationa d operational guide • Require financial i standards as a cond • Align enforcement a al legal clarity. • Incorporate technic ess testing methodo • Encourage industry- s, fintech licensin
	Financial Intelligence Units	• Financial Intelligence Units (FinCEN, FIU-IN	• [ASSESS]	• Investigate suspici ms, VASPs, or law e

		D, NCA, Egmont Group, etc/FATF)	Investigate blockchain-related threats to financial integrity and national security	• Trace on-chain fund ng technologies. • Assess links betwee raud rings, and ter • Validate and escala cial intelligence a • Coordinate domestic anisms and Egmont p
			• [ANALYZE] Monitor and interpret threat t rends across jurisdictions and technologies	• Aggregate and revie k patterns or techn • Analyze changes in ge hopping. • Correlate blockchai orks or jurisdictio • Track geographic tr ion. • Publish internal ri ers, and law enforc
			• [SHARE] Disseminate threat trend insig hts with domestic and global p artners	• Issue typology repo and threat patterns. • Contribute to Egmon • Host public-private ons and investigato

					• Facilitate bilatera y aligned with thre • Collaborate with te ven red flags.
				• [CONTRIBUTE] Inform international AML/CFT a nd virtual asset policy throug h threat-driven insights	• Provide anonymized lockchain services. • Propose revisions t nce with evolving t • Advise on enforceme digital assets. • Participate in FATF policy and financia • Support the harmoni dination and FATF e
				• [PROMOTE STANDARD] Recommend adoption of AML/CFT standards across blockchain st akeholders	• Advocate for integr of crypto-related b Recommend technical transaction monitor • Work with regulator ss. • Encourage cross-sec omply with AML/CFT • Support integration pliance teams.
		• Vulnerabil ity Coordi nation & D	• MITRE	• [WARN] Notify blockchain project team s of discovered vulnerabilitie	• Contact maintainers ilities or threat i

	Govt. Labs / Response Teams	isclosure	• national/institutional CERTs • National Labs	s or threats and recommend rem ediation	• Provide detailed te ioritize remediatio • Recommend timelines s. • Follow up on remedi • Where necessary, me icult to reach.
				• [DISCLOSE] Publicly disclose confirmed vu lnerabilities when remediation is complete or response is lac king	• Publish vulnerabili r a disclosure wind • Ensure disclosures • Collaborate with op • Balance transparenc high.
		• Cyber Thre at Modelin g & Intell igence		• [DISCLOSE] Share confirmed threat intelli gence with the public to raise ecosystem-wide awareness	• Publish technical r ampaigns, malware, • Share threat actor act manipulation st • Issue alerts and ad served threats. • Coordinate cross-se ess briefings.

		• Research & Tech Capability Development		• [ANALYZE] Conduct advanced testing, red teaming, and simulations to assess blockchain security	• Document proven defe • Publish open-source • Update public refere • Translate operationa bridges, L2s). • Apply frameworks li • Connect real-world • Update threat model
				• [SHARE] Share conducted research and developed tech capability	• Publish open-access n, or remediation s • Release open-source public research pro • Present findings at e practical adoptio • Collaborate with ac eview.
		• Public Infrastructure Support & Standards		• [CONTRIBUTE] Maintain a public repository of known vulnerabilities	• Operate or contribu s, consensus mechan • Organize vulnerabil ation and awareness. • Document incident r and regulators. • Maintain living rep on.

				• [CONTRIBUTE] Provide expert input into the development of technical and cybersecurity standards	• Participate in nati kchain security nor • Provide red-team-in onsensus validation • Contribute formal m • Act as conveners of in blockchain syste
				• [PROMOTE STANDARD] Build and deploy security tool s that reflect emerging standa rds and best practices	• Develop and publish stody systems, or v • Create automated to ation verifiers). • Partner with ecosys ts. • Promote adoption th
	Crypto-focused ISAC		• ISACs (CryptoISAC, JP-CryptoISAC, SEAL, etc)	• [DISCLOSE] Disseminate validated vulnerab ility and threat intelligence across the crypto sector	• Disseminate vetted artner organization • Standardize and dis t exploits) through • Coordinate timing o ppropriate. • Help anonymize sour • Provide summaries a g tools.
				• [ANALYZE]	• Review how leading practices.

			Evaluate remediation strategies and recommend practical responses for the industry	• Analyze the effecti s, frontend shutdow • Develop playbooks o nce attacks, bridge • Offer comparative a • Help translate comp s.
			• [SHARE] Circulate synthesized insights and post-incident learnings among ecosystem participants	• Publish threat tren • Facilitate closed-d efforts. • Share curated summa is. • Create anonymized i • Enable faster adopt • Share vulnerability
			• [CONTRIBUTE] Maintain curated repositories of blockchain-specific threat intelligence and security practices	• Host searchable dat s, techniques, proc • Aggregate and organ • Tag entries by asse bility. • Support structured nts. • Ensure repositories a trust framework.

			• [CONTRIBUTE] Support the development of blo ckchain-specific security and risk standards	• Facilitate working s operations, and inc • Provide recommendat • Coordinate communit • Advocate for intero s. • Support development ation.
			• [PROMOTE STANDARD] Encourage the adoption of reco gnized standards across the cr ypto industry	• Offer onboarding re • Develop tailored gu s, or custodial pro • Host compliance rea g resources. • Track and share ado • Incentivize conform or partner integrat
	ISACs for different industries	• ISACs for other indust ries (FS-ISAC, NH-ISA C, etc)	• [ANALYZE] Analyze cross-industry vulnera bility & threat intelligence	• Translate industry- • Distribute alerts a A, ENISA), and crit • Host multi-sector c cting multiple vert • Coordinate with sec across industries.
			• [SHARE] Relay relevant cross-sector vu lnerability & threat intellige nce to crypto-focused ISACs an d security coalitions	• Share anonymized in ant to blockchain s • Provide briefings o cidents (e.g., DDoS

				• Participate in join pto ISAC counterpar • Enable reciprocal a lti-industry situat • Help blockchain ISA o domain.
			• [CONTRIBUTE TO STANDARDS] Offer domain-specific expertis e to improve security and inte roperability standards	• Provide feedback on eir respective sect • Contribute insights esilience framework • Participate in mult finitions, metrics, • Recommend controls may apply to crypto • Facilitate dialogue interoperable stand
	International standards organi zation	• Technical standards or ganization (ISO) • Regulatory standards o rganization (FATF, IOS CO, FSB, etc) • Voluntary-based standa rds organization for c rypto (CCSS, BSSC, et c) • BGIN • OASIS	• [ANALYZE] Identify global patterns in vu lnerabilities and threats to i nform standard-setting	• Collect and synthes forcement, and indu • Analyze the impact oss blockchain syst • Monitor cross-borde n digital asset eco • Detect common chall into revision cycle • Integrate findings evelopment.
			• [SHARE]	• Publish reports, ty el weaknesses in bl

		• IETF	Disseminate threat and risk in sights to guide policy, compli ance, and technical adoption	• Host workshops, rou ng on emerging thre • Share anonymized ca ection, response, a • Provide practical g elopers adopting or • Issue updates and a ractices.
			• [STANDARDIZE] Develop internationally recogn ized frameworks for blockchain security and risk governance	• Facilitate multi-st cture, custody, key • Translate empirical d testing protocols. • Define taxonomies a sure a common langu • Develop modular, im open-source communi • Align blockchain-re ity standards used
			• [ANALYZE] Analyze trends on vulnerabilit ies and threats	• Monitor and assess feeds to identify c

	National Standards Organizatio n	• Technical standards or ganization (NIST and I PA) • Self-regulatory standa rds organizations		• Study incident case to national infrast • Identify systemic r es, custodial key m • Compare threat tren ameworks. • Integrate global th ector-specific need
			• [SHARE] Share analysis	• Publish risk assess g blockchain, finan • Host industry engag ion of security pra • Issue interpretive should apply emergi • Collaborate with go ld consensus around • Provide sector-spec to align blockchain
			• [STANDARDIZE] • Set national security guidelin es based on international stan dards	• Develop and publish ameworks such as IS • Tailor internationa gical maturity. • Create certificatio nce, healthcare, or • Provide normative l tional standards.

				• Coordinate with sec mpliance assessment